

HUY DUC (JUSTIN) VU

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SUMMARY

AI-focused Data Scientist and Engineer with hands-on experience building production-ready ML pipelines, RAG systems, and cloud data architectures. Proven track record of delivering measurable business impact by translating raw data into intelligent, scalable solutions. Currently advancing graduate-level expertise in Data Science at UTS, with a distinction average from a top-ranked application program.

EDUCATION

University of Technology Sydney	Feb 2025 – Dec 2026 (Expected)
<i>Master of Data Science and Innovation</i>	<i>GPA: 6.7/7</i>
University of Wollongong	Feb 2022 – Dec 2024
<i>Bachelor of Business, Business Analytics and Finance</i>	<i>WAM: 86/100</i>
Dean's Merit List 2022 & 2023; Ranked Top 2 Business Analytics Industry Project, Techniques and Tools in Business Analytics, Data Management and Security, and Corporate Finance; Top 3 in Big Data and Visual Analytics, Statistics, and Database Systems.	

SKILLS

- Programming language:** Python (Scikit-learn, TensorFlow, Pytorch, SpaCy, NLTK, Numpy, ...), SQL, R.
- AI/ ML:** OOP, NLP, LLM, VLM, RAG, LangChain, FAISS, Ollama.
- Cloud & Databases:** GCP, Azure, Snowflake, Databricks, Airflow, dbt, BigQuery, MySQL, PostgreSQL, MongoDB.
- Visualisation & Tools:** PowerBI, Tableau, SAS Viya, Excel, SPSS Statistics, Git, Docker, Claude Code, Cursor.

WORK EXPERIENCE

Vennu Australia <u>Business Data Analyst (Casual)</u>	Sydney, NSW Feb 2024 – Present
Vennu is a marketplace platform which lists spaces and connects with guests looking to book local, affordable venues for events.	
- Architected end-to-end ELT pipelines (Python, Google Maps API) integrating 10+ data sources into a unified Tableau dashboard, boosting strategic decision-making efficiency by 135% across 5+ stakeholder teams. - Drove a data-led marketing campaign by analysing customer engagement patterns, increasing social media reach by 120% and cutting the business planning cycle from 2 weeks to under 3 days (over 70% reduction). - Translated ambiguous stakeholder requirements into scalable data projects, iterating on prototypes through structured feedback loops to deliver production-ready analytics solutions.	
Vennu Australia <u>Business Analyst Intern</u>	Sydney, NSW Nov 2023 – Feb 2024
- Engineered 4 Power BI dashboards by extracting and transforming data from 6 MySQL sources, reducing reporting turnaround from 1–2 weeks to 2–3 days, a 70% time saving. - Produced a board-level proposal to adopt Power BI in daily operations, defining short- and long-term objectives; the initiative was approved and later scoped to include ML pipeline integration. - Delivered 2023 annual business review using Excel and Power BI, providing actionable insights adopted into the company's operational roadmap.	
UOW Chinese Academic Business Learning & Education (CABLE) <u>Senior Leader</u>	Wollongong, NSW Sep 2023 – Dec 2024
- Designed and facilitated Introduction to Statistics workshops for 50+ participants covering regression, probability distributions, and hypothesis testing, achieving a 99% satisfaction rate in post-session feedback. - Recognised as an academic high achiever within an AAUT award-winning program dedicated to outstanding student learning outcomes.	

PROJECTS

ELT Data Pipelines with Airbnb & Census Data (Airflow, dbt, Postgres, GCP)
- Engineered an automated Airflow ELT pipeline ingesting 300k+ Airbnb records and ABS Census data into a PostgreSQL data warehouse, enabling large-scale analytics across 128 NSW LGAs and 200+ suburbs - Designed a dbt medallion architecture (Bronze / Silver / Gold) to power revenue prediction models, host segmentation, and demographic profiling, identifying key drivers of listing performance and host financial viability.
Lecture Notes AI Tutor: Local RAG Assistant (RAG, LangChain, FAISS, Llama3, Ollama)
- Built a fully local, privacy-preserving RAG system enabling semantic search and grounded Q&A over university lecture PDFs, processing 1,300+ semantic chunks with optimised retrieval accuracy achieving 80% self-defined metrics. - Developed a complete data ingestion pipeline (PDF extraction, text cleaning, chunking, embedding) and led evaluation cycles to measurably improve answer faithfulness and relevance.